

Phase I/II Operating Characteristics

Cycle 1 futility cutoff comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
One-stage, futility = 0.95	27.2	15.1	2.4	0.1	0	55.2
One-stage, futility = 0.85	10.2	6.7	3.7	0.6	0.1	78.7
One-stage, futility = 0.75	2.6	2.9	2.3	0.4	0	91.8
Two-stage (highest utility), futility = 0.95	27.1	16	1.7	0.1	0.1	55.1
Two-stage (highest utility), futility = 0.85	10.3	8.5	3.3	0.6	0	77.3
Two-stage (highest utility), futility = 0.75	3.6	3.1	1.5	0.3	0	91.5
Two-stage (top-2 randomized), futility = 0.95	27.6	14	1.8	0.5	0.1	56
Two-stage (top-2 randomized), futility = 0.85	11.4	7.4	3	0.5	0.1	77.6
Two-stage (top-2 randomized), futility = 0.75	5.3	2.9	1.6	0.6	0.1	89.5
Mean patients treated						
One-stage, futility = 0.95	16.7	4.7	1.5	0.6	0.1	
One-stage, futility = 0.85	11.7	4.1	1.5	0.6	0.1	
One-stage, futility = 0.75	8.1	3.1	1.5	0.6	0.1	
Two-stage (highest utility), futility = 0.95	16.3	4.9	1.6	0.7	0.2	
Two-stage (highest utility), futility = 0.85	11.6	4.2	1.7	0.7	0.2	
Two-stage (highest utility), futility = 0.75	8	3.2	1.6	0.7	0.2	
Two-stage (top-2 randomized), futility = 0.95	16	5.1	1.6	0.6	0.1	
Two-stage (top-2 randomized), futility = 0.85	11.4	4.4	1.8	0.7	0.1	
Two-stage (top-2 randomized), futility = 0.75	8.1	3.3	1.7	0.7	0.1	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
One-stage, futility = 0.95	28.1	23.8	20.4	17.8	9.9	0
One-stage, futility = 0.85	29.4	23.1	20.8	16.9	9.8	0
One-stage, futility = 0.75	28	24	20.4	18	9.3	0.3
Two-stage (highest utility), futility = 0.95	29.5	23.1	19	17.6	10.7	0
Two-stage (highest utility), futility = 0.85	30.4	23	19.1	16.9	10.6	0
Two-stage (highest utility), futility = 0.75	28.1	24.3	18.2	18.4	10.6	0.4
Two-stage (top-2 randomized), futility = 0.95	34.2	22.7	16.5	15.5	11.1	0
Two-stage (top-2 randomized), futility = 0.85	32.5	25.4	15.8	16.4	9.8	0.1
Two-stage (top-2 randomized), futility = 0.75	33.2	24.1	16.3	16.1	9.8	0.5
Mean patients treated						
One-stage, futility = 0.95	8	6.8	5.8	5.4	4.1	
One-stage, futility = 0.85	7.9	6.9	5.7	5.4	4.1	
One-stage, futility = 0.75	7.9	6.9	5.7	5.4	4.1	
Two-stage (highest utility), futility = 0.95	7.7	6.8	5.7	5.4	4.3	
Two-stage (highest utility), futility = 0.85	7.8	6.8	5.7	5.4	4.3	
Two-stage (highest utility), futility = 0.75	7.6	6.9	5.7	5.4	4.3	
Two-stage (top-2 randomized), futility = 0.95	7.8	6.8	5.6	5.5	4.3	
Two-stage (top-2 randomized), futility = 0.85	7.7	6.9	5.6	5.6	4.2	
Two-stage (top-2 randomized), futility = 0.75	7.6	6.9	5.6	5.6	4.3	

All rows use the shared-alpha additive model, continuous enrollment, and toxicity/efficacy IPDE alpha = 0.

Utility 3 scores: no toxicity/no efficacy = 40; no toxicity/efficacy = 100; toxicity/no efficacy = 0; toxicity/efficacy = 60. Stop % is no-OBD selection.

Phase I/II Operating Characteristics

Cycle 1 futility cutoff comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
One-stage, futility = 0.95	10.2	51.7	22.9	11.7	2.3	1.1
One-stage, futility = 0.85	8.6	50.6	23.4	11	2.6	3.8
One-stage, futility = 0.75	6	43.7	24.3	11.2	2.2	12.6
Two-stage (highest utility), futility = 0.95	11.1	52.8	23.1	9.9	2.1	1
Two-stage (highest utility), futility = 0.85	8.8	51.7	24.6	9.5	2.1	3.3
Two-stage (highest utility), futility = 0.75	6.9	46.1	22.8	9.5	2	12.7
Two-stage (top-2 randomized), futility = 0.95	9.2	56	21.1	11.2	2.3	0.3
Two-stage (top-2 randomized), futility = 0.85	7.5	55.2	22.5	10	1.8	3
Two-stage (top-2 randomized), futility = 0.75	5.3	48.9	21.2	11.3	2	11.3
Mean patients treated						
One-stage, futility = 0.95	6.7	11.2	6.7	3.9	1.3	
One-stage, futility = 0.85	6.3	11.2	6.7	3.9	1.3	
One-stage, futility = 0.75	5.7	10.6	6.6	3.9	1.3	
Two-stage (highest utility), futility = 0.95	6.3	10.7	7.2	4.2	1.5	
Two-stage (highest utility), futility = 0.85	6	10.6	7.2	4.2	1.5	
Two-stage (highest utility), futility = 0.75	5.4	10.1	7	4.1	1.5	
Two-stage (top-2 randomized), futility = 0.95	7.6	9.8	7	4.2	1.4	
Two-stage (top-2 randomized), futility = 0.85	7.1	9.9	7.1	4.2	1.4	
Two-stage (top-2 randomized), futility = 0.75	6.2	9.4	7	4.1	1.4	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
One-stage, futility = 0.95	1.5	2	6	58.2	32.3	0
One-stage, futility = 0.85	1.4	2.1	5.7	59.2	31.6	0
One-stage, futility = 0.75	1.4	1.3	5.7	58.7	32.8	0.1
Two-stage (highest utility), futility = 0.95	1.9	1.5	6.6	55	35.1	0
Two-stage (highest utility), futility = 0.85	1.7	1.1	5.7	56.6	34.9	0
Two-stage (highest utility), futility = 0.75	1.5	1.5	6	56	34.9	0.1
Two-stage (top-2 randomized), futility = 0.95	0.4	1.1	3	54.4	41	0
Two-stage (top-2 randomized), futility = 0.85	0.3	0.7	3.5	54.4	41.1	0
Two-stage (top-2 randomized), futility = 0.75	0.4	0.7	3.7	54.8	40.4	0
Mean patients treated						
One-stage, futility = 0.95	3.6	3.5	4.4	12.3	6.2	
One-stage, futility = 0.85	3.5	3.5	4.4	12.4	6.1	
One-stage, futility = 0.75	3.5	3.5	4.4	12.4	6.1	
Two-stage (highest utility), futility = 0.95	3.5	3.5	4.3	10.8	8	
Two-stage (highest utility), futility = 0.85	3.5	3.5	4.2	10.8	7.9	
Two-stage (highest utility), futility = 0.75	3.5	3.5	4.2	10.8	7.9	
Two-stage (top-2 randomized), futility = 0.95	4	4	4.8	9.1	8.1	
Two-stage (top-2 randomized), futility = 0.85	4	4	4.8	9.2	8	
Two-stage (top-2 randomized), futility = 0.75	3.9	4	4.8	9.2	8	

All rows use the shared-alpha additive model, continuous enrollment, and toxicity/efficacy IPDE alpha = 0.

Utility 3 scores: no toxicity/no efficacy = 40; no toxicity/efficacy = 100; toxicity/no efficacy = 0; toxicity/efficacy = 60. Stop % is no-OBD selection.

Phase I/II Operating Characteristics

Cycle 1 futility cutoff comparison - N = 30, Cycle_Max = 1

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
One-stage, futility = 0.95	18.1	27.2	42	10.5	0.8	1.3
One-stage, futility = 0.85	14.1	25.3	42	10.6	0.8	7.2
One-stage, futility = 0.75	9.8	20.9	40.8	10	0.8	17.7
Two-stage (highest utility), futility = 0.95	18.7	25.7	43.7	9.4	1.1	1.5
Two-stage (highest utility), futility = 0.85	14	25.1	43.9	9.4	1	6.6
Two-stage (highest utility), futility = 0.75	10.3	20.4	40.6	9.1	1.1	18.5
Two-stage (top-2 randomized), futility = 0.95	15.3	27.6	43	12	1	1
Two-stage (top-2 randomized), futility = 0.85	12.2	26.8	42.5	11.3	1.1	6.1
Two-stage (top-2 randomized), futility = 0.75	8.5	21.7	41	11.2	0.8	16.8
Mean patients treated						
One-stage, futility = 0.95	8.8	7.8	9.3	3.4	0.6	
One-stage, futility = 0.85	8.1	7.7	9.2	3.4	0.6	
One-stage, futility = 0.75	7.2	7.1	9	3.3	0.6	
Two-stage (highest utility), futility = 0.95	8.2	7.7	9.3	3.7	0.9	
Two-stage (highest utility), futility = 0.85	7.5	7.7	9.3	3.7	0.9	
Two-stage (highest utility), futility = 0.75	6.7	7	8.8	3.6	0.9	
Two-stage (top-2 randomized), futility = 0.95	9.2	8.3	7.9	3.6	0.9	
Two-stage (top-2 randomized), futility = 0.85	8.5	8.3	7.9	3.6	0.9	
Two-stage (top-2 randomized), futility = 0.75	7.5	7.6	7.7	3.6	0.9	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
Utility 3	41	40.8	42.4	43	58	
OBD selection (%)						
One-stage, futility = 0.95	12.2	11.7	8	8.1	59.6	0.3
One-stage, futility = 0.85	9.4	10.7	10.7	8.3	59.5	1.4
One-stage, futility = 0.75	5.8	7.7	9.3	11	59.6	6.6
Two-stage (highest utility), futility = 0.95	12.3	12.4	9.2	7.8	58	0.3
Two-stage (highest utility), futility = 0.85	9.7	11.3	11.5	8.7	57.8	1
Two-stage (highest utility), futility = 0.75	7.5	8.2	8.9	10.6	58.4	6.4
Two-stage (top-2 randomized), futility = 0.95	13.8	11.7	7.1	8	59.3	0.1
Two-stage (top-2 randomized), futility = 0.85	10.2	9.8	9.8	9.8	59.2	1.2
Two-stage (top-2 randomized), futility = 0.75	7.2	8.7	8.3	9.9	59.8	6.1
Mean patients treated						
One-stage, futility = 0.95	5.9	5.2	4.5	4.8	9.6	
One-stage, futility = 0.85	5.6	5.2	4.6	4.8	9.6	
One-stage, futility = 0.75	5.2	4.8	4.7	4.9	9.6	
Two-stage (highest utility), futility = 0.95	5.7	5.1	4.4	4.9	9.7	
Two-stage (highest utility), futility = 0.85	5.5	5.1	4.5	4.9	9.8	
Two-stage (highest utility), futility = 0.75	5.1	4.8	4.6	4.9	9.8	
Two-stage (top-2 randomized), futility = 0.95	7	6.1	4.8	5	7.1	
Two-stage (top-2 randomized), futility = 0.85	6.6	6.1	5.1	5.1	7.1	
Two-stage (top-2 randomized), futility = 0.75	6	5.7	5.1	5.2	7.1	